Day 2

NCHC X OpenACC X NVIDIA

OpenACC
More Science, Less Programming

2026 NCHC Open Hackathon

Important Dates

- ~~10 / 14 (Wed)： Day 0: Kick-off Meeting 15:00~17:00 PM (Online)~~
- ~~10 / 21 (Wed)： Day 1: Scrum #1 Meeting 15:00~16:00 PM (Online)~~
- **10 / 28 (Wed)： Day 2: Scrum #2 Meeting 15:00~16:00 PM (Online)**
  - 15:00 - 15:10 : All-hand meeting
  - 15:10 - 16:00 : Group Scrum Meeting
    - 5~10 mins for each team
- 11 / 04 (Wed)： Final Day 09:30~14:00 PM (In-person at NCHC)

跨領域加速進展 TRACK A

科學運算・量子・資料中心 ｜ 起始／量測基準 → 最新回報

實測／限定工作量　跨硬體・多 GPU　預估　待對照／待補量測

各列工作量不同，不宜排名或取平均

團隊	應用領域	Baseline → Latest		加速／狀態	比較條件與驗證範圍
A1-01 **Team NTU**	氣象／多相流模擬	grind time	0.1149 → **0.0496** ns	**2.32× †** 跨報告換算	同規模測例、8×H200；需同次複測確認。
A1-02 **The Quantum Annealers**	量子啟發最佳化 金融詐欺特徵選擇	同一新測例	1 CPU core → **RTX 5090**	**26.2×** 暖機後求解器	新測例對照，非 Day 0 累積；解品質各有勝負。
A1-03 **西灣海環 • SCHISM**	海流預報／HPC	180 steps	180.1 → **83.6** s	**2.15×** 同節點時間步進	相同4個Grace節點；通過96小時容差驗收。
A1-04 **open_qblock**	量子校準／AI代理	collection	64.4 → **4.5** s	**約14×** 每輪校準耗時	主要來自重置／批次；量子品質另行驗證。
A2-05 **TreeLine**	DPU／DDoS偵測	插入10K規則	10.7 → **3.8** ms	**約2.8×** 規則插入／跨世代	BF-2→BF-3；非吞吐加速，100G掃描待測。
A2-06 **Zero-Wait Prefix**	DPU／LLM服務	暖請求TTFT	77.8 → **63.5*** ms	**預估1.23×** 尚非端到端實測	假設session猜對；未計scatter成本。
A2-07 **Lab 117**	DPU／RDMA多播	1→3複製	無對照值 → **92.6 Gbps**	**待基準** 已展示新能力	三接收端總輸出；不是每端速率或3倍加速。

† 衍生比值或跨資源比較，詳見註記；* 為預估，非已達成結果。無對照值不硬填倍速。

Day 0 未提供數值或設定不同時，採最早可比量測／原管線重測；來源為隊伍投影片，非獨立驗證。

跨領域加速進展 TRACK B

醫療・天文・農業・語音・機器人 | 起始／量測基準 → 最新回報

實測／限定工作量　跨硬體・多 GPU　預估　待對照／待補量測

各列工作量不同，不宜排名或取平均

團隊	應用領域	Baseline → Latest	加速／狀態	比較條件與驗證範圍
B3-08 **BioVibe-Oncology**	生醫／癌症抗藥性	Chai-1基準　39 h → **未對齊重測**	**待對照** 最新：Day 2	跨模型約9／26／84秒；品質未對齊，非加速倍數。
B3-09 **PairLab**	多模態AI／血液病理	DS-DFCPF　158 → **17** min VLM-MIL　500 → **42** min	**9.3× / 11.9× †** 兩條研究管線	VLM：1×3090→5×H200；含平行化效益。
B3-10 **3T→7T MRI**	醫療影像／MRI合成	訓練　2.56 → **1.82** h 推論　41.35 → **26.78** s	**1.41× / 1.54×** 訓練／推論	H200新版對照；不與Day 0舊架構直接相除。
B3-11 **Lab210**	醫療影像／心臟CT問答	32切片訓練　40 h 40 m → **20 h 36 m**	**1.97× †** 跨硬體完整epoch	2×L40S→1×H200；品質待評估，非128切片。
B4-12 **Talk2ALMA**	天文／資料檢索	啟動首結果　190 → **7.5** s 每次查詢　10.6 → **3.9** s	**約25× / 2.7×** 啟動／暖查詢	不同指標，不相乘；不混入跨硬體訓練比較。
B4-13 **CropSight**	農業／作物影像辨識	每批32張　1,306 → **504** ms	**2.59×** 同批次管線	採Day2重測基準；準確度尚待回報。
B4-14 **NTU_SPML**	語音／口語語言模型	8×H200　82 → **1,157** samples/s	**14.1×** 樣本吞吐量	高LR仍有發散；最佳設定完整訓練待驗證。
B4-15 **HSCC**	視覺語言／影片事件定位	bench100　188–189 → **47–48** min	**約4.0×** 單GPU完整流程	同1×H200；僅100影片，完整測試待完成。
B4-16 **Parallel Robotics**	機器人／四足強化學習	PhysX→Newton　31 → **20** h	**約1.55×** 30k迭代耗時	相同L40配置；大規模平均接近，尾端仍有差距。

† 衍生比值或跨資源比較，詳見註記；* 為預估，非已達成結果。無對照值不硬填倍速。

Day 0 未提供數值或設定不同時，採最早可比量測／原管線重測；來源為隊伍投影片，非獨立驗證。

Nov 12 Final Day Lunchbox and Attendance

[2026 NCHC Open Hackathon - Lunchbox Survey – Fill out form]

All attendees are required, including mentors, please make sure reply it by this Wed, Oct 28, thanks.





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- **11 / 04 (Wed)：Final Day 09:30~14:00 PM (In-person at NCHC)**
  - 09:30~10:00: Opening & Meet up
  - 10:00~12:30: Final Presentation
  - 12:30~13:00: Wrap-up & Closing
  - 13:00~14:00 : Lunch together ~

NCHC NVIDIA OPEN HACKATHON 2026

FINAL PRESENTATION AGENDA

Wednesday, November 4, 2026 · Asia/Taipei

09:30–10:00 | **Opening & Meet up**

10:00–12:30 | **Final Presentation · Parallel Tracks A & B** | 15 minutes per team · 12-min presentation + 3-min Q&A

TRACK A

Host: Jay Chen

10:00–10:15	A1-01	Team NTU
10:15–10:30	A1-02	The Quantum Annealers
10:30–10:45	A1-03	Baytech · SCHISM
10:45–11:00	A1-04	open_qblock
11:00–11:15	A2-06	DPU KV-cache Prefetch
11:15–11:30	A2-07	Lab 117
11:30–11:45	A2-05	TreeLine
11:45–12:30		Discussion / buffer

TRACK B

Host: Yang-Hsien Lin

10:00–10:15	B3-08	BioVibe-Oncology
10:15–10:30	B3-09	Pair Lab
10:30–10:45	B3-10	3T-to-7T MRI
10:45–11:00	B3-11	Lab210
11:00–11:15	B4-12	Talk2ALMA
11:15–11:30	B4-13	CropSight
11:30–11:45	B4-14	NTU_SPML
11:45–12:00	B4-15	HSCC
12:00–12:15	B4-16	Parallel Robotics
12:15–12:30		Discussion / buffer

12:30–13:00 | **Wrap-up & Closing**

13:00–14:00 | **Lunch together ~**

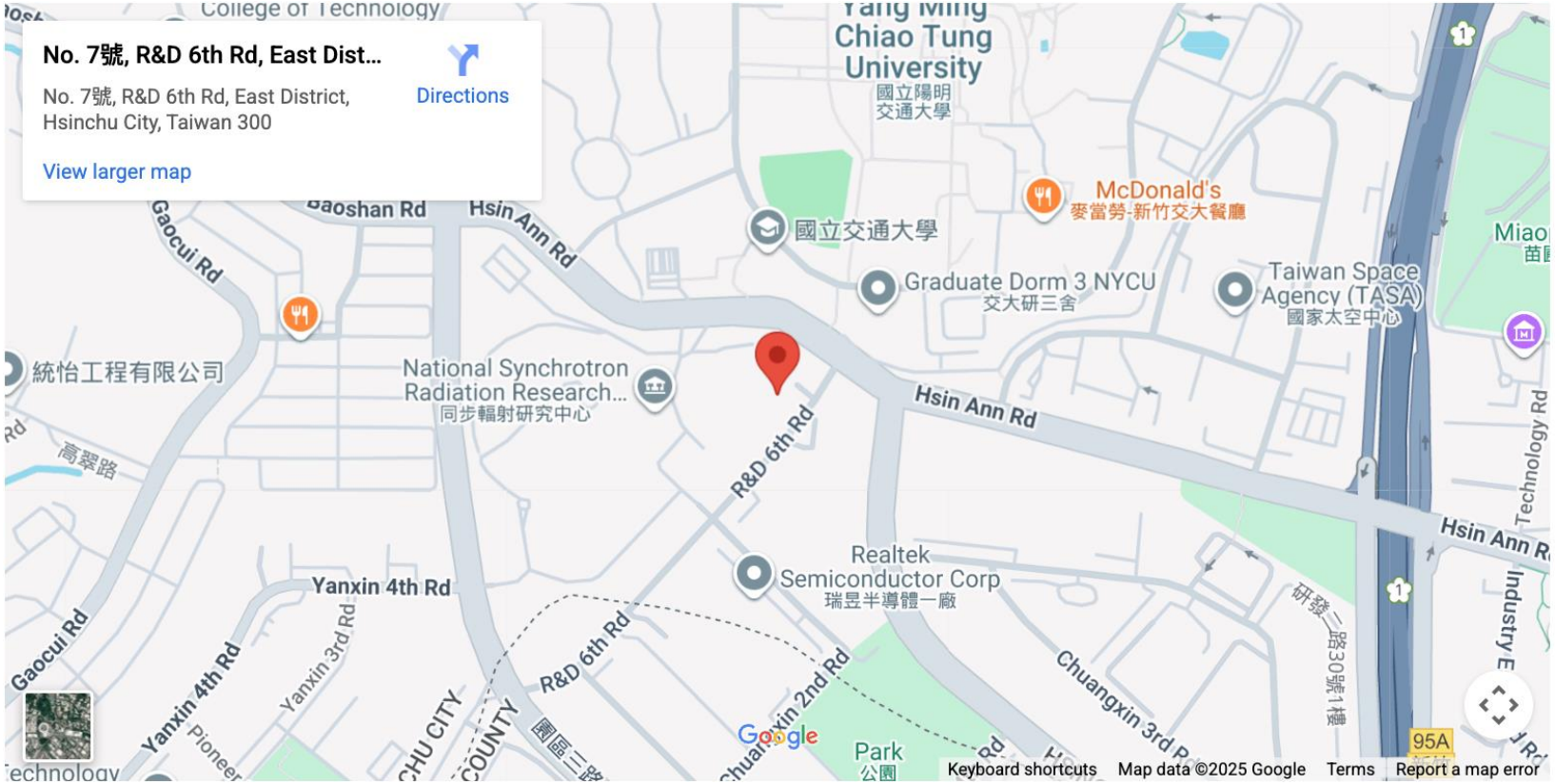
OpenACC More Science, Less Programming · OPEN HACKATHONS · NCHC 國家實驗研究院 國家高速網路與計算中心 National Center for High-performance Computing · NVIDIA

Transportation to NCHC Hsinchu Office

https://www.nchc.org.tw/Page?itemid=85&mid=100

Address: No. 7, R&D 6th Rd., East Dist., Hsinchu City, Taiwan, R.O.C. 30076

Hsinchu traffic during rush hour can be quite heavy. Please allow extra travel time and ensure you arrive by **9:20 AM**.



Parking


Click the button and say hi, the administrator will open the gate.
Parking Area
P

Presentation Room

Every presenter will use a clip-on microphone to record the voice.



Presentation

- All presentations will be recorded and published on NCHC YouTube Channel, please let us know if any confidential information.
- A kind reminder: the next presenting team should move to the front row.
- 12 Minutes Presentation + 3 Minutes Q&A.

- Small gifts will be awarded to the highest-voted team in each track during the final presentations:
  - 4 x NVIDIA logo USB-C Cable
  - 4 x NVIDIA logo Laptop Tote
  - 1 x NVIDIA logo Playing Cards and Pins









OpenACC More Science, Less Programming
OPEN HACKATHONS
NCHC 國家實驗研究院 國家高速網路與計算中心 National Center for High-performance Computing
NVIDIA

Day Final - Template

OpenACC
More Science, Less Programming

Total presentation time is 12 minutes

Team Name

Team Members (Name, organization, and picture)
Team Mentors (Name, organization, and picture)

Application Name

- Problem the team is trying to solve.
- Scientific driver for the chosen algorithm.
- What's the algorithmic motif?
- What parts are you focused on?

Evolution and Strategy

- What was your goal for coming here?
- What was your initial strategy?
- How did this strategy change?

Results and Final Profile

- What were you able to accomplish?
- Did you achieve a speed up?
- Show multi-core vs. GPU numbers
- What did you learn?
- Did you create a new algorithm?
- Did you achieve new scientific goals?

Energy Efficiency

INPUTS	
# CPU Cores	64
# GPUs (A100)	6
Application Speedup	20.0x

Node Replacement	13.3x

GPU NODE POWER SAVINGS			
	AMD Dual Rome 7742	8x A100 80GB SXM4	Power Savings
Compute Power (W)	14,667	6,500	8,167
Networking Power (W)	619	93	526
Total Power (W)	**15,286**	**6,593**	8,693

Node Power efficiency	2.3x

ANNUAL ENERGY SAVINGS PER GPU NODE			
	AMD Dual Rome 7742	8x A100 80GB SXM4	Power Savings
Compute Power (kWh/year)	128,480	56,940	71,540
Networking Power (kWh/year)	5,424	814	4,610
Total Power (kWh/year)	**133,904**	**57,754**	**76,150**

$/kWh	$	0.18
Annual Cost Savings	$	13,707.04
3-year Cost Savings	$	41,121.13

Metric Tons of CO2	54
Gasoline Cars Driven for 1 year	12
Seedlings Trees grown for 10 years	892

(source: Link)

The calculator will compare the consumption of a number of CPU-only nodes with dual CPUs required to perform the same amount of work as 1 GPU node with 2 CPUs and 8 GPUs.

1. Use this calculator for your report
2. Add your acceleration numbers in the INPUTS section
3. Modify $/kwh number if necessary
4. Paste a screenshot similar to the one on the right in this slide to report energy efficiency of your project

What problems have you encountered?

- Problems with legacy app structure.
- Issues with algorithms.
- Tool bugs.
- Tool lacking features.
- System setup.

Wishlist

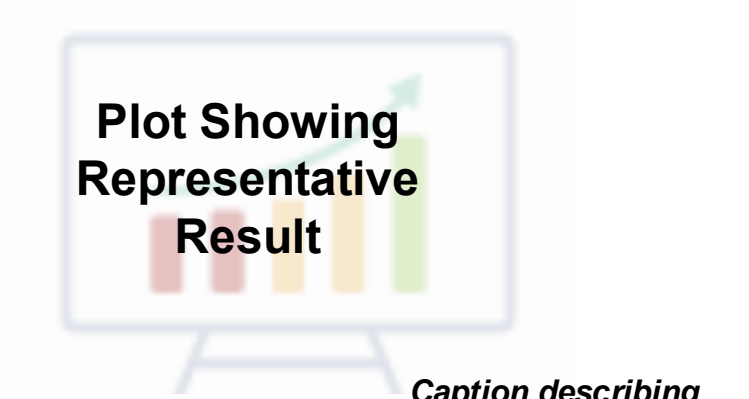
- What do you wish existed to make your life easier?
- Tools
- Language standards
- Event
- Systems

Final Thoughts

- Was this Open Hackathon worth it?
- Will you continue development?
- Next steps, future plans.
- What sustained resources or support will be critical for your work after the event?

Application Background

High-level description of application and uses.
Light on domain-specific jargon; should be appropriate for a general technical audience.
Targeted computational motifs.



Caption describing figure in simple terms

Hackathon Objectives and Approach

Programming models.
Profiling/hot spots
Refactoring
Libraries
Performance tuning
Other

Technical Accomplishments and Impact

What were you able to achieve at the hackathon?
How did you achieve it?
Speedup
Why does it matter/what does it enable?

淺顯易懂的標題

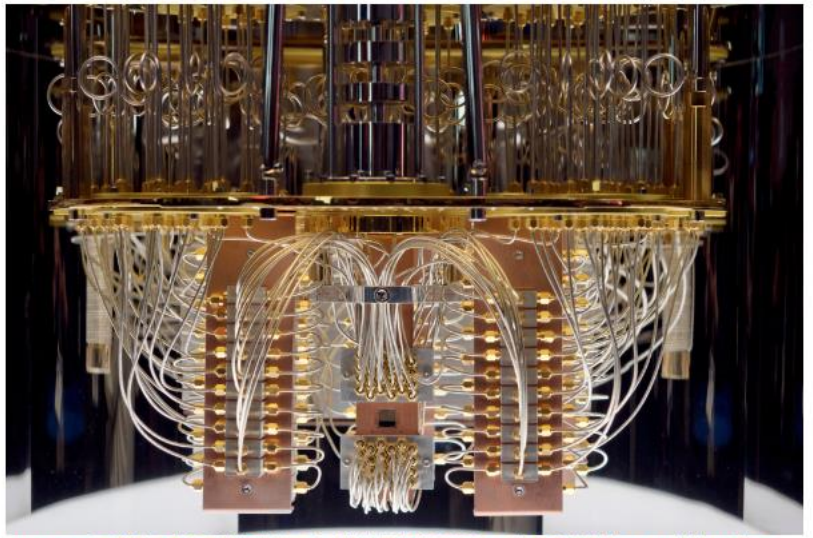
研究領域示意圖

{__} 團隊來自 {__}老師帶領的{__}實驗室，
AI Agent 將 {__} 工作流程加速了 {__} 倍；
CPU/GPU/DPU 將 {__} 研究加速了 {__} 倍！！

250~500 字儘量易懂的描述。
為什麼這個領域重要？為什麼這個命題重要？
加速成果？加速帶來什麼影響和潛力？

和加速成果相關的補充數據

量子算法模擬



haofan2023團隊成員來自臺灣大學資工系「洪士灝老師實驗室」，將量子演算法QAOA加速468倍！
— NVIDIA Mentors: Tian Zheng, Frank Lin, Yun-Yuan Wang

量子技術正以驚人的速度發展，預示著我們即將進入量子計算的時代。在這個過程中，量子電路模擬成為一個關鍵工具，它在量子硬體和軟體的開發中扮演著重要的角色，特別是在處理量子程式的編寫和驗證方面。傳統電腦的強模擬能夠獲得完整的量子狀態信息。這使得傳統電腦在構建量子系統方面變得不可或缺，尤其是在當前噪聲較多的中等規模量子（NISQ）時代。

量子近似優化算法（QAOA）是一種常用的量子算法，用於通過近似解來解決組合優化問題。然而，在虛擬量子計算機上執行QAOA對於解決需要大規模量子電路模擬的組合優化問題而言，會遇到模擬速度較慢的問題。團隊使用數學優化來壓縮量子操作，並結合有效的位元操作進一步降低計算複雜性，透過GPU加速最高獲取468倍的加速效果！

Table 1: The elapsed time of 5-level QAOA (unit: second, double).

Qubit	CPU_{Single}	$CPU_{Mutiple}$	CPU_{Cache}	GPU_{Cache}	GPU_{All}
23	29.80	1.28 (23x)	1.28 (63x)	0.24 (120x)	0.06 (**341x**)
24	68.00	3.46 (20x)	3.46 (43x)	0.55 (123x)	0.12 (**382x**)
25	152.52	15.32 (10x)	15.31 (45x)	1.19 (127x)	0.23 (**404x**)
26	330.69	33.83 (10x)	33.83 (56x)	2.60 (126x)	0.56 (**417x**)
27	712.26	72.66 (10x)	72.66 (54x)	5.59 (127x)	1.08 (**427x**)
28	1556.87	156.52 (10x)	156.52 (54x)	11.96 (130x)	2.17 (**445x**)
29	3325.55	335.09 (10x)	335.09 (49x)	25.73 (129x)	4.45 (**451x**)
30	7226.46	718.33 (10x)	718.33 (47x)	55.20 (130x)	9.22 (**468x**)

更多資訊請看：https://github.com/nqobu/nvidia/raw/main/20231207/Team02.pdf

2024 Final Report Examples

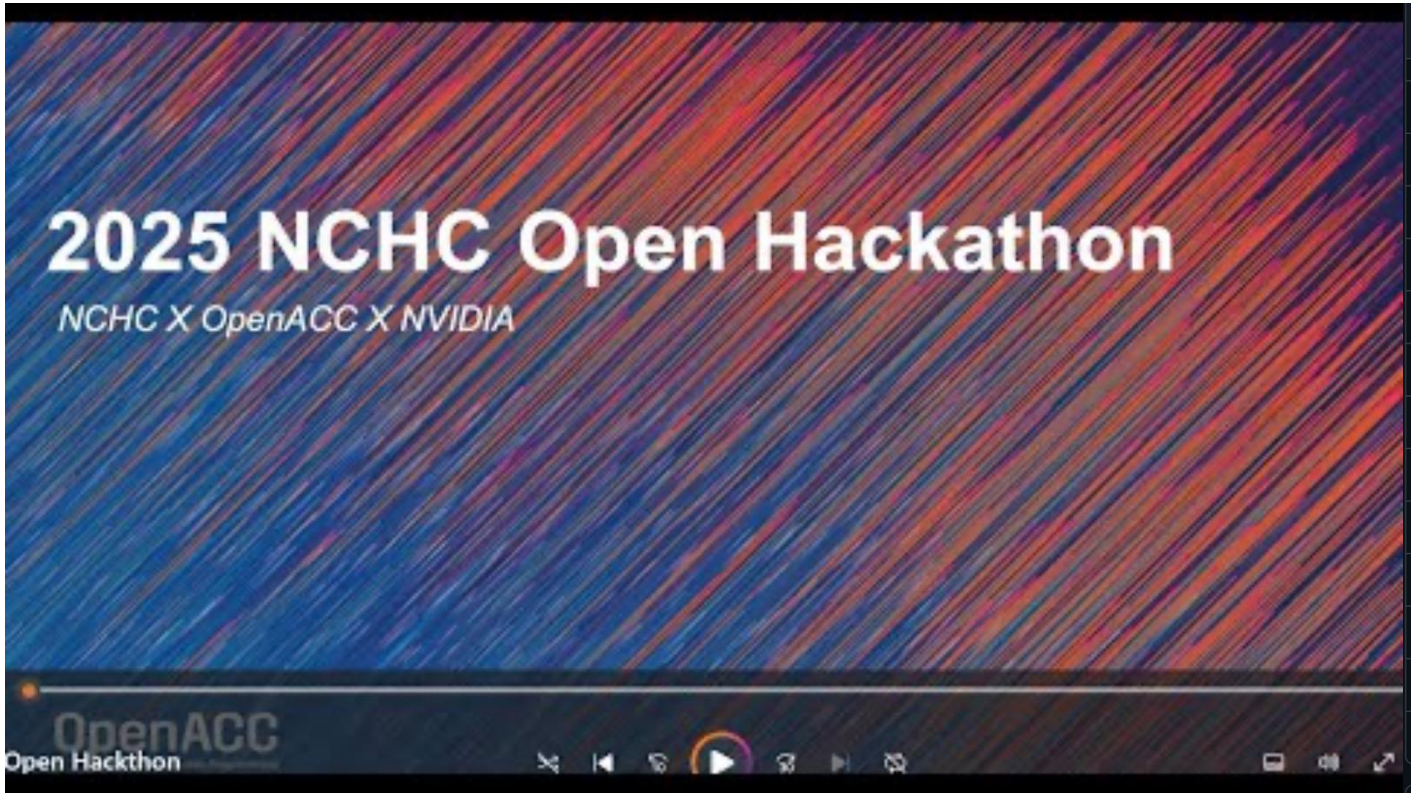
https://github.com/nqobu/nvidia/tree/main/20241204
https://www.youtube.com/watch?v=Ahf9sp3hCpc

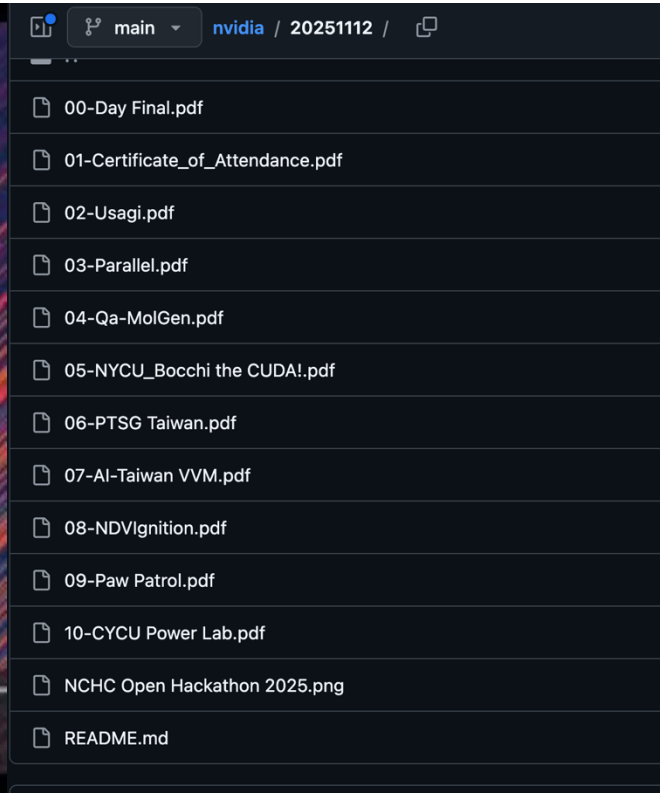


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11-Plantmen.pdf	Add files via upload	10 months ago
12-CYCU_Quantum.pdf	Add files via upload	10 months ago

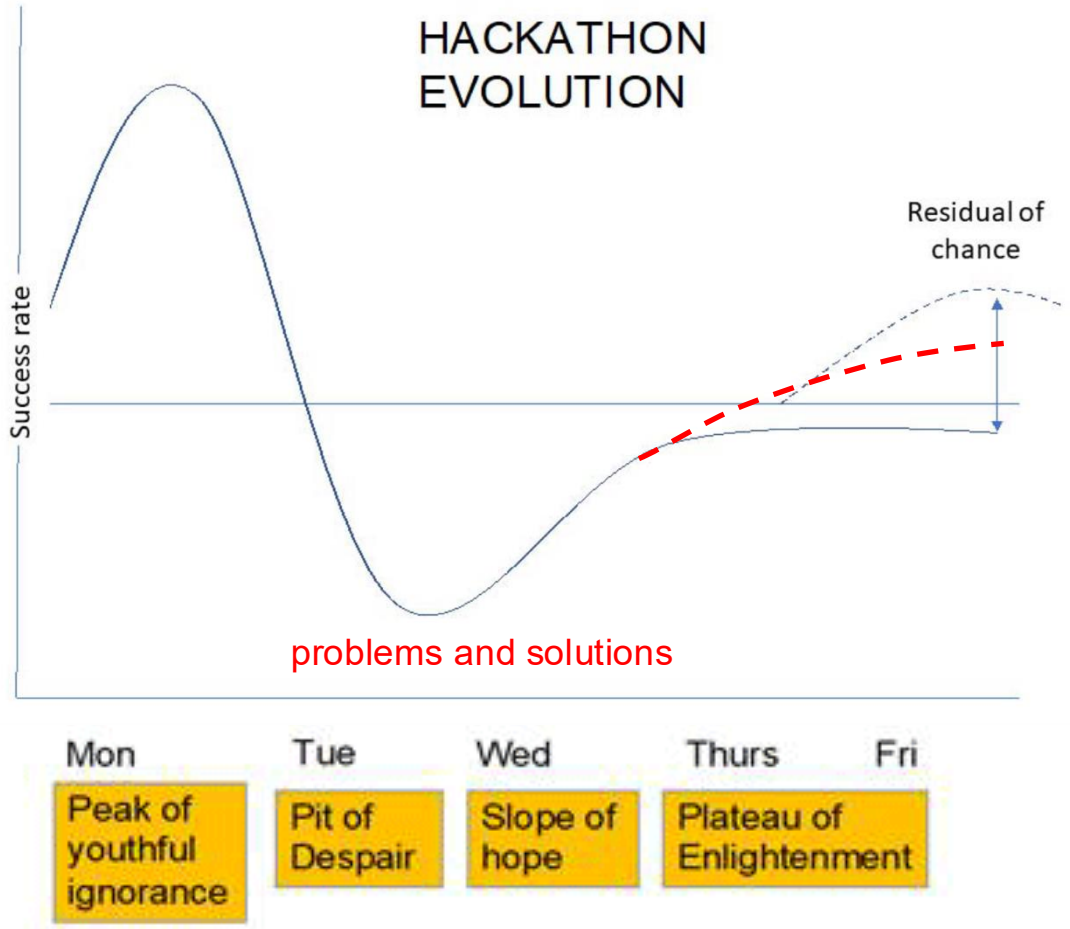
2025 Final Report Examples

https://github.com/nqobu/nvidia/tree/main/20251112
https://www.youtube.com/watch?v=cuMgrmRtOKE





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This article highlights the Oak Ridge Leadership Compute Facilitys GPU Hackathon
https://www.computer.org/csdl/magazine/cs/2018/04/mcs2018040095/13rRUxYIN88